AiPERF

InfluxDB Export, Restore and Streamlit Setup Guide

Windows | InfluxDB OSS 1.8.10 | Local Demo Environment

Purpose Copy the AiPERF jmeter database to another Windows machine, restore it safely as jmeter_aiperf_demo, configure the AiPERF repository, and run the Streamlit user interface.

Document type	Technical setup SOP
Source platform	AiPERF
Database	InfluxDB OSS 1.8.10 / jmeter
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1. Scope and Recommended Topology

This guide uses a portable InfluxDB backup to create an independent local demonstration environment. The friend's machine receives a point-in-time copy of the database and does not need a live network connection to the source machine.

```
Source machine
  InfluxDB OSS 1.8.10: jmeter
       | portable backup + ZIP
       v
Approved transfer channel
       |
       v
Friend machine
  InfluxDB OSS 1.8.10: jmeter_aiperf_demo
  AiPERF Git repository + local .env
  Streamlit UI: http://localhost:8501
```

Security Share the database backup and sanitized configuration only. Do not share personal API keys, Jenkins credentials, GitHub tokens, passwords, or the source .env file.

2. Prerequisites and Transfer Package

- Windows PowerShell on both machines.
- InfluxDB OSS 1.8.10 Windows AMD64 on both machines.
- Python and Git on the friend's machine.
- Authorized access to the sureshaiperf/AiPERF repository.
- Approved internal transfer channel such as SharePoint, OneDrive, or Teams.

Files to provide to the friend

- aiperf_jmeter_backup.zip
- Access to the AiPERF Git repository
- A sanitized .env.example or the environment-variable list in this guide
- Separate approved LLM and embedding credentials only when dynamic AI functions are required

3. Export the jmeter Database on the Source Machine

3.1 Define paths

```
$InfluxHome = "C:\Coforge Software\influxdb-1.8.10_windows_amd64\influxdb-1.8.10-1"
$BackupRoot = "C:\aiperf_db_backup"
$BackupPath = "$BackupRoot\jmeter_portable"
$ZipPath = "$BackupRoot\aiperf_jmeter_backup.zip"
```

3.2 Verify version and database

```
& "$InfluxHome\influxd.exe" version

& "$InfluxHome\influx.exe" `
    -host "127.0.0.1" `
    -port 8086 `
    -execute "SHOW DATABASES"
```

Confirm that the output contains the database name jmeter.

3.3 Verify the backup service

```
Test-NetConnection 127.0.0.1 -Port 8088
```

Expected result: TcpTestSucceeded : True. Port 8086 is the normal HTTP/query endpoint; port 8088 is used by the InfluxDB 1.x backup and restore service.

3.4 Create a clean backup directory

```
if (Test-Path $BackupPath) {
    Remove-Item $BackupPath -Recurse -Force
}

New-Item -ItemType Directory -Path $BackupPath -Force | Out-Null
```

3.5 Create the portable backup

```
& "$InfluxHome\influxd.exe" backup `
    -portable `
    -database "jmeter" `
    -host "127.0.0.1:8088" `
    "$BackupPath"
```

Compatibility fallback If the executable rejects -database, repeat the command using -db "jmeter" instead.

```
& "$InfluxHome\influxd.exe" backup `
    -portable `
    -db "jmeter" `
    -host "127.0.0.1:8088" `
    "$BackupPath"
```

3.6 Verify and compress

```
Get-ChildItem $BackupPath |
    Select-Object Name, Length, LastWriteTime

$Files = Get-ChildItem $BackupPath -File
if ($Files.Count -eq 0) {
    throw "Backup folder is empty. Do not transfer this backup."
}
Write-Host "Backup verification passed. Files found: $($Files.Count)" -ForegroundColor Green
```

A successful portable backup normally contains .manifest, .meta and .tar.gz files.

```
if (Test-Path $ZipPath) {
    Remove-Item $ZipPath -Force
}

Compress-Archive `
    -Path "$BackupPath\*" `
    -DestinationPath "$ZipPath" `
    -CompressionLevel Optimal

Get-Item $ZipPath |
    Select-Object FullName, Length, LastWriteTime
```

Transfer: C:\aiperf_db_backup\aiperf_jmeter_backup.zip

4. Prepare the Friend's Machine

4.1 Verify Python and Git

```
python --version
git --version

# If python is not recognized:
py --version
```

4.2 Install or extract InfluxDB OSS 1.8.10

Recommended installation directory:

```
C:\Coforge Software\influxdb-1.8.10_windows_amd64\influxdb-1.8.10-1

$InfluxHome = "C:\Coforge Software\influxdb-1.8.10_windows_amd64\influxdb-1.8.10-1"
& "$InfluxHome\influxd.exe" version
```

4.3 Create local data directories

```
$InfluxDataRoot = "C:\influxdb-data"
New-Item -ItemType Directory -Path "$InfluxDataRoot\meta" -Force | Out-Null
New-Item -ItemType Directory -Path "$InfluxDataRoot\data" -Force | Out-Null
New-Item -ItemType Directory -Path "$InfluxDataRoot\wal" -Force | Out-Null
```

4.4 Generate and update influxdb.conf

```
& "$InfluxHome\influxd.exe" config > "$InfluxHome\influxdb.conf"
notepad "$InfluxHome\influxdb.conf"
```

Confirm or update the following entries. Use forward slashes in TOML paths:

```
bind-address = "127.0.0.1:8088"

[meta]
  dir = "C:/influxdb-data/meta"

[data]
  dir = "C:/influxdb-data/data"
  wal-dir = "C:/influxdb-data/wal"

[http]
  enabled = true
  bind-address = ":8086"
  auth-enabled = false
```

4.5 Start InfluxDB

Keep PowerShell window 1 open:

```
& "$InfluxHome\influxd.exe" `
    -config "$InfluxHome\influxdb.conf"
```

In PowerShell window 2, verify both ports:

```
Test-NetConnection 127.0.0.1 -Port 8086
Test-NetConnection 127.0.0.1 -Port 8088
```

5. Restore the Database on the Friend's Machine

5.1 Extract the transferred ZIP

```
$BackupZip = "C:\Temp\aiperf_jmeter_backup.zip"
$RestorePath = "C:\Temp\aiperf_jmeter_backup"

if (Test-Path $RestorePath) {
    Remove-Item $RestorePath -Recurse -Force
}
New-Item -ItemType Directory -Path $RestorePath -Force | Out-Null
Expand-Archive -Path $BackupZip -DestinationPath $RestorePath -Force
Get-ChildItem $RestorePath -Recurse | Select-Object FullName, Length
```

5.2 Check existing databases

```
& "$InfluxHome\influx.exe" `
    -host "127.0.0.1" `
    -port 8086 `
    -execute "SHOW DATABASES"
```

5.3 Restore into a safe demo database

```
& "$InfluxHome\influxd.exe" restore `
    -portable `
    -host "127.0.0.1:8088" `
    -db "jmeter" `
    -newdb "jmeter_aiperf_demo" `
    "$RestorePath"
```

Why a new name? jmeter_aiperf_demo avoids overwriting or conflicting with an existing local jmeter database.

5.4 Resolve a “manifest not found” error

```
Get-ChildItem $RestorePath -Filter "*.manifest" -Recurse |
    Select-Object -ExpandProperty DirectoryName -Unique
```

Use the returned directory as the final path in the restore command.

6. Validate the Restored Database

```
& "$InfluxHome\influx.exe" `
    -host "127.0.0.1" `
    -port 8086 `
    -execute "SHOW DATABASES"

& "$InfluxHome\influx.exe" `
    -host "127.0.0.1" `
    -port 8086 `
    -database "jmeter_aiperf_demo" `
    -execute "SHOW MEASUREMENTS"
```

Verify representative AiPERF measurements

```
& "$InfluxHome\influx.exe" -host "127.0.0.1" -port 8086 `
    -database "jmeter_aiperf_demo" `
    -execute 'SELECT * FROM "aiperf_execution_fingerprint" ORDER BY time DESC LIMIT 5'

& "$InfluxHome\influx.exe" -host "127.0.0.1" -port 8086 `
    -database "jmeter_aiperf_demo" `
    -execute 'SELECT * FROM "aiperf_service_metrics" ORDER BY time DESC LIMIT 5'
```

If a sample measurement is absent, use SHOW MEASUREMENTS to identify the exact names that exist in the backup. The absence of one optional measurement does not by itself indicate a failed restore.

7. Clone and Prepare AiPERF

7.1 Clone the repository

```
New-Item -ItemType Directory -Path "C:\practice" -Force | Out-Null
Set-Location "C:\practice"

git clone https://github.com/sureshaiperf/AiPERF.git
Set-Location "C:\practice\AiPERF"
```

7.2 Create and activate a virtual environment

```
python -m venv .venv
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1
python -c "import sys; print(sys.executable)"
```

7.3 Install dependencies

```
python -m pip install --upgrade pip setuptools wheel
python -m pip install -r "C:\practice\AiPERF\requirements.txt"
python -m streamlit version
```

Fallback only If Streamlit is not included in the repository requirements, install the minimum runtime packages with: python -m pip install streamlit python-dotenv influxdb pandas numpy requests

7.4 Verify core modules

```
python -c "import streamlit; print('Streamlit:', streamlit.__version__)"
python -c "import influxdb; print('InfluxDB client imported successfully')"
python -c "import dotenv; print('python-dotenv imported successfully')"
```

8. Configure the Local .env

Create C:\practice\AiPERF\.env or copy the repository .env.example. Do not commit .env to Git.

```
INFLUXDB_HOST=127.0.0.1
INFLUXDB_PORT=8086
INFLUXDB_DATABASE=jmeter_aiperf_demo
INFLUXDB_USERNAME=
INFLUXDB_PASSWORD=

AIPERF_INFLUXDB_HOST=127.0.0.1
AIPERF_INFLUXDB_PORT=8086
AIPERF_INFLUXDB_DATABASE=jmeter_aiperf_demo
AIPERF_INFLUXDB_USERNAME=
AIPERF_INFLUXDB_PASSWORD=
```

The duplicate variable families are a temporary compatibility safeguard if older and newer AiPERF modules use different configuration names.

Optional AI configuration

```
PRIMARY_API_URL=<approved-primary-model-endpoint>
PRIMARY_API_KEY=<friend-specific-approved-key>
PRIMARY_MODEL=<approved-model-name>

FAILOVER_API_URL=<approved-failover-endpoint>
FAILOVER_API_KEY=<friend-specific-approved-key>
FAILOVER_MODEL=<approved-failover-model>

EMBEDDING_API_URL=<approved-embedding-endpoint>
EMBEDDING_API_KEY=<friend-specific-approved-key>
AIPERF_EMBEDDING_DIMENSIONS=736
AIPERF_EMBEDDING_MAX_CHARS=<repository-approved-value>
EMBEDDING_TIMEOUT_SECONDS=<repository-approved-value>
```

Offline UI behavior Without LLM credentials, restored metrics and charts can still work. Dynamic GPT answers, embedding generation, and semantic retrieval require approved endpoint access and credentials.

9. Run the AiPERF Streamlit UI

9.1 Find the Streamlit entry point

```
Get-ChildItem "C:\practice\AiPERF" -Filter "aiperf_ui.py" -Recurse |
    Select-Object -ExpandProperty FullName
```

9.2 Recommended startup command

```
Set-Location "C:\practice\AiPERF"
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1

python -m streamlit run ".\baselineintelligence\aiperf_ui.py"
```

Open http://localhost:8501 in a browser.

Alternative port

```
python -m streamlit run ".\baselineintelligence\aiperf_ui.py" `
    --server.port 8502
```

Trusted local-network access

```
python -m streamlit run ".\baselineintelligence\aiperf_ui.py" `
    --server.address "0.0.0.0" `
    --server.port 8501
```

Only use this configuration on a trusted network. Do not expose Streamlit directly to the public internet without authentication, TLS, and an approved reverse proxy.

10. Optional One-Command Streamlit Start Script

Save the following as C:\practice\AiPERF\start_aiperf_ui.ps1:

```
$ErrorActionPreference = "Stop"

$RepoRoot = "C:\practice\AiPERF"
$VirtualEnvironment = Join-Path $RepoRoot ".venv"
$StreamlitApp = Join-Path $RepoRoot "baselineintelligence\aiperf_ui.py"
$InfluxUrl = "http://127.0.0.1:8086/ping"

if (-not (Test-Path $RepoRoot)) { throw "AiPERF repository not found: $RepoRoot" }
if (-not (Test-Path "$VirtualEnvironment\Scripts\python.exe")) { throw "Virtual environment not found:
$VirtualEnvironment" }
if (-not (Test-Path $StreamlitApp)) { throw "Streamlit application not found: $StreamlitApp" }

try {
    Invoke-WebRequest -Uri $InfluxUrl -Method Get -UseBasicParsing -TimeoutSec 5 | Out-Null
}
catch {
    $StatusCode = $_.Exception.Response.StatusCode.value__
    if ($StatusCode -ne 204) { throw "InfluxDB is not reachable at $InfluxUrl" }
}

Set-Location $RepoRoot
Write-Host "Starting AiPERF Streamlit UI..." -ForegroundColor Green
& "$VirtualEnvironment\Scripts\python.exe" -m streamlit run "$StreamlitApp" `
    --server.address "127.0.0.1" --server.port 8501
```

Run the script:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
& "C:\practice\AiPERF\start_aiperf_ui.ps1"
```

11. Troubleshooting

Symptom	Likely cause	Resolution
“unknown arguments: backup ...”	The command used influx.exe instead of influxd.exe.	Run backup and restore with influxd.exe. Use influx.exe only for query commands.
Port 8088 is closed	Backup service bind address is not active or InfluxDB is not running.	Set bind-address = "127.0.0.1:8088" in influxdb.conf, restart InfluxDB, and retest.
No manifest found	The ZIP created an extra folder level.	Search recursively for *.manifest and use its parent directory as the restore path.
No module named streamlit	Streamlit is not installed in the active virtual environment.	Use .venv\Scripts\python.exe -m pip install streamlit, then run with python -m streamlit.
No module named streamlit.proto	The Streamlit installation is damaged or mismatched.	Uninstall and force reinstall Streamlit inside the AiPERF virtual environment.
UI opens but shows no data	The .env database name or host does not match the restored database.	Verify jmeter_aiperf_demo, run SHOW MEASUREMENTS, restart Streamlit after editing .env.
Environment variables do not load	The UI resolves .env relative to a different working directory.	Start from the repository root. If required temporarily, place a sanitized .env beside aiperf_ui.py.

12. Daily Startup Sequence

PowerShell window 1: InfluxDB

```
$InfluxHome = "C:\Coforge Software\influxdb-1.8.10_windows_amd64\influxdb-1.8.10-1"
& "$InfluxHome\influxd.exe" -config "$InfluxHome\influxdb.conf"
```

PowerShell window 2: AiPERF Streamlit

```
Set-Location "C:\practice\AiPERF"
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1
python -m streamlit run ".\baselineintelligence\aiperf_ui.py"
```

Browser

```
http://localhost:8501
```

13. Acceptance Checklist

- ☐ InfluxDB version is 1.8.10 on the friend's machine.
- ☐ Ports 8086 and 8088 are reachable locally.
- ☐ jmeter_aiperf_demo appears in SHOW DATABASES.
- ☐ SHOW MEASUREMENTS returns AiPERF measurements.
- ☐ The AiPERF virtual environment is active.
- ☐ python -m streamlit version succeeds.
- ☐ .env points to 127.0.0.1:8086 and jmeter_aiperf_demo.
- ☐ The UI loads at http://localhost:8501.
- ☐ Historical statistics are visible.
- ☐ LLM functions are either working with approved credentials or clearly marked unavailable.

Operational limitation The restored database is a point-in-time snapshot. New Jenkins executions on the source machine will not automatically synchronize. Create a new backup when refreshed demo data is required, or later implement an approved centralized database architecture.